

Space group 166 as maximal subgroup of space group 224

The purpose of these notes is to show that space group 224 reduces to space group 166 when its nonsymmorphic symmetries are broken. It's not hard (can be verified directly in e.g. `Crystalline.jl`) to see that the symmetry operations coincide. The more difficult thing is to see that the lattice vectors indeed coincide. The lattice basis of the conventional cell of space group 166 is given by (for a free parameter, α ; also we use p, c superscripts to indicate quantities corresponding to conventional and primitive lattices, respectively):

$$\mathbf{R}_1^{c,166} = (1, 0, 0), \mathbf{R}_2^{c,166} = (-1/2, \sqrt{3}/2, 0), \mathbf{R}_3^{c,166} = (0, 0, \alpha)$$

The primitive lattice, on the other hand is spanned by the following primitive lattice vectors:

$$\mathbf{R}_1^{p,166} = \frac{1}{2}(1, 1/\sqrt{3}, 2\alpha/3), \mathbf{R}_2^{p,166} = \frac{1}{2}(-1, 1/\sqrt{3}, 2, 2\alpha/3), \mathbf{R}_3^{p,166} = \frac{1}{2}(0, -2/\sqrt{3}, 2\alpha/3)$$

Of course, the conventional lattice vectors must be expressible in terms of integer linear combinations of the primitive lattice vectors. Indeed, we have:

$$\begin{aligned} \mathbf{R}_1^c &= \mathbf{R}_1^{p,166} - \mathbf{R}_2^{p,166} \\ \mathbf{R}_2^c &= \mathbf{R}_2^{p,166} - \mathbf{R}_3^{p,166} \\ \mathbf{R}_3^{c,166} &= \mathbf{R}_1^{p,166} + \mathbf{R}_2^{p,166} + \mathbf{R}_3^{p,166} \end{aligned}$$

Similarly, we may write the primitive lattice in terms of the conventional lattice:

$$\begin{aligned} \mathbf{R}_1^{p,166} &= 2/3\mathbf{R}_1^{c,166} + 1/3\mathbf{R}_2^{c,166} + 1/3\mathbf{R}_3^{c,166} \\ \mathbf{R}_2^{p,166} &= -1/3\mathbf{R}_1^{c,166} + 1/3\mathbf{R}_2^{c,166} + 1/3\mathbf{R}_3^{c,166} \\ \mathbf{R}_3^{p,166} &= -1/3\mathbf{R}_1^{c,166} - 2/3\mathbf{R}_2^{c,166} + 1/3\mathbf{R}_3^{c,166} \end{aligned}$$

Note that we may thus write all of the group elements of space group 166 as the symmorphic parts plus or minus $\mathbf{R}_1^{p,166}$. In the appendix, we show how one may verify that the symmorphic part of the spacegroup operations (in the basis of the conventional unit cell) may be written in the basis of the primitive unit cell.

In order to see that the primitive lattice may correspond to a cubic lattice (i.e. that it could correspond to a subgroup of spacegroup 224), we must judiciously choose the free parameter, α . We choose this parameter in such a way that the primitive lattice vectors are mutually orthogonal:

$$\mathbf{R}_1^{p,166} \cdot \mathbf{R}_2^{p,166} = \frac{1}{4} \left[-1 + \frac{1}{3} + \frac{4\alpha^2}{9} \right] \rightarrow \alpha = \sqrt{3/2}, \alpha/3 = 1/\sqrt{6}$$

Note that this choice for α also makes the second and third primitive lattice vectors and first and third primitive lattice vectors orthogonal:

$$\mathbf{R}_2^{p,166} \cdot \mathbf{R}_3^{p,166} = \mathbf{R}_1^{p,166} \cdot \mathbf{R}_3^{p,166} = \frac{1}{4} \left[-\frac{2}{3} + \frac{4\alpha^2}{9} \right] = 0$$

Thus, a choice of the free parameter, α , given by $\alpha = \sqrt{3/2}$ gives a cubic primitive lattice.

Appendix: Verifying the consistency between group operations in the conventional and primitive bases

Using `Crystalline.jl`, we may verify the consistency of the group operations in the primitive and conventional bases as follows:

```
sgnum = 166
num_symmorphic = 12
dbasis = directbasis(sgnum)
pbasis = primitivize(dbasis, centering(sgnum))
transformation_matrix = inv(hcat(dbasis...)) * hcat(pbasis...) # Transforms a vector
# conventional basis
sg = spacegroup(sgnum)
ops_conventional = sg[1:num_symmorphic]
ops_primitive = primitivize(sg)

function p_to_c(x)
    return inv(transformation_matrix)*x*transformation_matrix
end

for (op_conventional, op_primitive) in zip(ops_conventional, ops_primitive)
    op_conventional_pbasis = p_to_c(op_conventional.rotation)
    @assert (round.(op_conventional_pbasis, digits=5) == op_primitive.rotation)
end
```